

Daljeet Singh

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FORMAL EDUCATION

Amity University, Mumbai

Panvel, Navi Mumbai

Bachelor of Science (B.Sc.) Information Technology, Secondary Field in French and Political Science July. 2025

- **Relevant Coursework:** Software Engineering, Public Speaking, Data Structures with C++, Operating Systems, Introduction to Machine Learning, Business Intelligence, Cloud Computing (Azure & AWS), Java, DBMS

EXPERIENCES

Digital Systems & Growth Lead

Sept. 2023 – Sept. 2024

Multi-Channel Portfolio

Mumbai, MH

- Scaled a multi-channel digital portfolio to **2M+** aggregate views by leveraging data-driven SEO, metadata schemas, and targeted technical content deployment.
- Leveraged Generative AI for automated asset production across both platforms, reducing time-to-market for new content by **50%**

PROJECTS

daljeetsingh.me/#/projects

Janus | Golang, IP Blocking, Docker, AI Scraper Mitigation, TLS Fingerprinting, HTML — 🌀 Aug. 2024 - Sept. 2025

Bot Mitigation & Traffic Analysis Engine

- Built a system that **checks multiple security signals** at once - TLS connection patterns, geographic location risks, unique browser fingerprints, and request frequency — to accurately identify automated bots.
- Created a challenge that **requires light computation from suspicious visitors**, issues a secure session token upon success, and keeps verification under 0.1 seconds for real users to prevent friction
- Designed a **RESTful API** with two primary endpoints: one to submit client fingerprints, and another to request and return challenge verification, ensuring straightforward integration.

Hull Tactical Market Prediction | LightGBM, Polars, Scikit-learn, Joblib, ONNX, PyTorch — 🌀 Sept – Dec. 2025

Developed an ensemble machine learning system to predict S&P 500 excess returns.

- Developed a fast **data processing pipeline using Polars** that creates features in **real-time from over 200 market indicators** across seven financial categories.
- Built a **memory-efficient rolling buffer using deque structures** that stores one year of historical data, ensuring time-based predictions remain sequential and prevent future data contamination.

RESEARCH WORK

Research Publication:

June. 2024 – Present

“Enhancing Accessibility through Real-Time Scene Understanding & Navigation for Visually Impaired”

First Author, Published on TechRxiv.org (Preprint Server) | DOI: [10.22541/au.176843166.60388360/v1](https://doi.org/10.22541/au.176843166.60388360/v1)

Co-authored with Dr. Narayan Kulkarni, Assistant Professor, Amity Institute of Information Technology (AIIT)

- Developed a **dual-stream deep learning framework** combining SSD-VGG16 and CRNN for real-time object detection and text recognition, **optimized for resource-constrained** mobile devices.
- Validated system performance on 1,000+ real-world images across diverse lighting and environmental conditions, demonstrating **practical applicability for assistive navigation**.

SKILLS

Languages: Python, Golang, SQL, Bash, C/C++, Java, HTML/CSS

Tools/Frameworks: Django, Fast API, Svelte, Postman, Git

Databases: PostgreSQL, MongoDB

DevOps Tools: Linux, Nginx, Docker, Cloudflare, GitHub Actions